

Q5B: Calculate the relativistic momentum and energy of a particle undergoing acceleration.

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Q5B: Relativistic Momentum and Energy of an Accelerating Particle

Setup

Consider a particle of rest mass m_0 undergoing **constant proper acceleration** α , starting from rest at $t = 0$ in the lab frame S . Proper acceleration is the acceleration measured in the instantaneous rest frame of the particle, and is *not* equivalent to a constant lab-frame force.

Proper Acceleration: Correct Definition

The 4-acceleration is defined as:

$$A^\mu = \frac{dU^\mu}{d\tau}$$

where $U^\mu = \gamma(c, \mathbf{v})$ is the 4-velocity and τ is proper time.

The magnitude of the 4-acceleration is the proper acceleration α :

$$A^\mu A_\mu = -\alpha^2 \quad (\text{using signature } +---)$$

In 1+1 dimensions this gives:

$$\gamma^4 \left(\frac{dv}{dt} \right)^2 \cdot \frac{1}{c^2} \cdot c^2 = \alpha^2 \implies \gamma^3 \frac{dv}{dt} = \alpha$$

Hence the lab-frame force is *not* constant:

$$F = \frac{dp}{dt} = \frac{d(\gamma m_0 v)}{dt} = m_0 \gamma^3 \frac{dv}{dt} \cdot \frac{1}{\gamma^2}$$

More precisely, from $p = \gamma m_0 v$ and the chain rule:

$$\frac{dp}{dt} = m_0 \gamma^3 \frac{dv}{dt} = m_0 \alpha \gamma^2$$

This is velocity-dependent and **not constant**, unlike the previous (incorrect) treatment.

Velocity as a Function of Coordinate Time

We integrate the proper acceleration equation. Since $\gamma^3 \frac{dv}{dt} = \alpha$, write:

$$\frac{dv}{dt} = \frac{\alpha}{\gamma^3} = \alpha \left(1 - \frac{v^2}{c^2} \right)^{3/2}$$

Separating variables:

$$\int_0^v \frac{dv'}{(1 - v'^2/c^2)^{3/2}} = \alpha t$$

The left-hand side evaluates to:

$$\frac{v}{\sqrt{1 - v^2/c^2}} = \alpha t$$

Solving for $v(t)$:

$$v(t) = \frac{\alpha t}{\sqrt{1 + \frac{\alpha^2 t^2}{c^2}}}$$

As $t \rightarrow \infty$, $v(t) \rightarrow c$, confirming the speed of light is never reached.

Lorentz Factor

$$\gamma(t) = \frac{1}{\sqrt{1 - v^2/c^2}} = \frac{1}{\sqrt{1 - \frac{\alpha^2 t^2}{c^2 + \alpha^2 t^2}}} = \sqrt{\frac{c^2 + \alpha^2 t^2}{c^2}}$$

$$\gamma(t) = \sqrt{1 + \frac{\alpha^2 t^2}{c^2}}$$

Relativistic Momentum

$$p(t) = \gamma(t) m_0 v(t) = \sqrt{1 + \frac{\alpha^2 t^2}{c^2}} \cdot m_0 \cdot \frac{\alpha t}{\sqrt{1 + \alpha^2 t^2/c^2}}$$

$$p(t) = m_0 \alpha t$$

Note: although $p(t) = m_0 \alpha t$ resembles the Newtonian result, it arises here from integrating the proper-acceleration equation of motion, *not* from a constant force assumption. The lab-frame force $F = dp/dt = m_0 \alpha$ happens to be constant as a consequence of the algebra, but the physical origin is proper acceleration, not a prescribed constant force.

Relativistic Total Energy

$$E(t) = \gamma(t) m_0 c^2 = m_0 c^2 \sqrt{1 + \frac{\alpha^2 t^2}{c^2}}$$

Relativistic Kinetic Energy

$$K(t) = E(t) - m_0 c^2 = m_0 c^2 \left(\sqrt{1 + \frac{\alpha^2 t^2}{c^2}} - 1 \right)$$

Proper Time

Integrating $d\tau = dt/\gamma(t)$:

$$\tau(t) = \int_0^t \frac{dt'}{\gamma(t')} = \int_0^t \frac{dt'}{\sqrt{1 + \alpha^2 t'^2/c^2}} = \frac{c}{\alpha} \sinh^{-1} \left(\frac{\alpha t}{c} \right)$$

Inverting: $t = \frac{c}{\alpha} \sinh \left(\frac{\alpha \tau}{c} \right)$, so the worldline in terms of proper time is:

$$x(\tau) = \frac{c^2}{\alpha} \left(\cosh \frac{\alpha \tau}{c} - 1 \right), \quad t(\tau) = \frac{c}{\alpha} \sinh \frac{\alpha \tau}{c}$$

which is the standard **hyperbolic motion** of uniformly accelerated relativistic particles.

Energy–Momentum Relation Verification

$$E^2 - (pc)^2 = m_0^2 c^4 \left(1 + \frac{\alpha^2 t^2}{c^2}\right) - m_0^2 \alpha^2 t^2 c^2 = m_0^2 c^4 \quad \checkmark$$

Classical Limit

For $\alpha t \ll c$:

$$\gamma(t) \approx 1 + \frac{\alpha^2 t^2}{2c^2}, \quad v(t) \approx \alpha t, \quad K(t) \approx \frac{1}{2} m_0 \alpha^2 t^2 = \frac{1}{2} m_0 v^2 \quad \checkmark$$

Numerical Example

Let $m_0 = 1.673 \times 10^{-27}$ kg (proton), $\alpha = 10^{16}$ m/s², $t = 10^{-8}$ s.

$$\alpha t = 10^8 \text{ m/s}, \quad \frac{\alpha t}{c} = \frac{10^8}{3 \times 10^8} = \frac{1}{3}$$

$$\gamma = \sqrt{1 + \frac{1}{9}} = \sqrt{\frac{10}{9}} \approx 1.054$$

$$p = m_0 \alpha t = (1.673 \times 10^{-27})(10^8) = 1.673 \times 10^{-19} \text{ kg m/s}$$

$$E = \gamma m_0 c^2 \approx 1.054 \times (1.673 \times 10^{-27})(9 \times 10^{16}) \approx 1.586 \times 10^{-10} \text{ J}$$

$$K = (\gamma - 1)m_0 c^2 \approx 0.054 \times 1.506 \times 10^{-10} \approx 8.13 \times 10^{-12} \text{ J}$$

$$\tau = \frac{c}{\alpha} \sinh^{-1}\left(\frac{1}{3}\right) = \frac{3 \times 10^8}{10^{16}} \sinh^{-1}(0.333) \approx 3 \times 10^{-8} \times 0.3272 \approx 9.82 \times 10^{-9} \text{ s}$$

Summary Table

Quantity	Expression (constant proper acceleration α)
Velocity	$v(t) = \frac{\alpha t}{\sqrt{1 + \alpha^2 t^2 / c^2}}$
Lorentz factor	$\gamma(t) = \sqrt{1 + \frac{\alpha^2 t^2}{c^2}}$
Relativistic momentum	$p(t) = m_0 \alpha t$
Total energy	$E(t) = m_0 c^2 \sqrt{1 + \frac{\alpha^2 t^2}{c^2}}$
Kinetic energy	$K(t) = m_0 c^2 \left(\sqrt{1 + \frac{\alpha^2 t^2}{c^2}} - 1 \right)$
Proper time	$\tau = \frac{c}{\alpha} \sinh^{-1}\left(\frac{\alpha t}{c}\right)$